Question #1 of 18

Question ID: 1587542

The $250 million endowment of Stevenson University (SU) has a target 4% spending rate to support research at SU. The endowment is currently invested as follows: 30% domestic equities, 20% foreign equities, 20% domestic fixed income, and 30% foreign fixed income. Last month, the endowment received a $150 million donation from a philanthropist—the largest ever in its history. Which of the following approaches by SU endowment's portfolio managers would be *most appropriate*?

A) Invest the additional $150 million in riskier assets to increase the potential of higher portfolio returns. ✗

B) Invest the additional $150 million using the same asset allocation and weights as the existing portfolio, in line with the portfolio's strategic asset allocation. ✗

C) Evaluate opportunities of adding new asset classes and changing the asset class weights on the combined $400 million portfolio. ✓

Explanation

With the additional donation, the endowment's size increased materially (i.e., 60%). Therefore, it allows the portfolio managers to take a holistic view of the portfolio by exploring new asset classes that may have been unavailable under the previous portfolio size as well as to reevaluate the portfolio's asset allocation, liquidity, and time horizon.

Investing the donation simply using the existing asset allocation ignores the potential for new asset classes and the revised asset set and risk opportunities. Investing the donation simply in riskier assets ignores the existing assets and fails to take a holistic view of the portfolio.

(Module 5.2, LOS 5.c)

Question #2 of 18

Question ID: 1551742

If an investment is held in a tax-exempt account, then the investor bears:

A) all of the investment risk. ✓

B) some of the investment risk. ✗

C) none of the investment risk. ✗

Explanation

In a taxable account, losses realized result in a reduction in taxes that serve to offset the magnitude of the loss. Thus, some of the downside risk is transferred to the government. In a tax-exempt account, the variability of returns is not affected by the taxes.

(Module 5.1, LOS 5.b)

Question #3 of 18

Question ID: 1587541

The Howarth School Foundation (HSF) has historically provided 30% of the annual operating costs of running the Howarth law school. Recent changes in government funding mean that previously fully funded scholarships will now only be 40% funded, and that the HSF must provide more support to the school. Based on this information, the HSF will *most likely*:

A) increase the allocation to riskier timber and farmland to generate higher expected return and cover the required outflows. ✗

B) decrease the allocation to private equity to increase the liquidity of the portfolio. ✓

C) increase the allocation to zero-coupon bonds to increase duration. ✗

Explanation

The school is becoming more dependent on the foundation, decreasing the ability to take risk and increasing liquidity demands for payouts to the school. Shifting to more liquid investments makes sense. Higher duration is not relevant and is probably riskier—plus, zero-coupons provide no regular cash flow. Higher return may seem logical, but taking more risk to get it is not appropriate when the ability to take risk has declined—plus, land is an illiquid asset.

(Module 5.2, LOS 5.c)

Question #4 of 18

Question ID: 1587534

Global Auto Ltd. (GAL) maintains a fully funded $500 million pension fund for its defined benefit pension plan that is now closed to new employees. GAL's retirement age is 62, and the average age of the company's pension plan beneficiaries is 55. However, there is an upcoming change in pension legislation that will allow employees to retire earlier, at age 58. The pension fund is currently invested as follows:

- 20% domestic equities
- 20% foreign equities
- 30% domestic bonds with a duration of 15 years
- 30% alternative investments (equally invested in real estate, hedge funds, and private equity)

The *most appropriate* strategy by GAL's pension fund manager would be to reduce the allocation to:

A) alternative investments and increase the allocation to domestic equities.

B) domestic and foreign equities and increase the allocation to fixed income with a shorter duration.

C) alternative investments and equities and increase the allocation to fixed income with a shorter duration.

Explanation

Although the pension plan is currently fully funded, it has a high exposure to potentially illiquid alternative investments. In addition, the duration of its fixed-income portfolio is too high and exceeds the average time until retirement (7 years, assuming age 62 retirement age). Furthermore, the reduction in the average retirement age to 58 will result in increased liquidly needs. The fund needs to reduce its allocation to riskier assets (alternative investments and equity) and increase its allocation to less risky assets (fixed income with a lower duration).

(Module 5.1, LOS 5.a)

Question #5 of 18

Question ID: 1587540

Gil Tabor, CFA, and Jan Sills, CFA, are discussing how the choice of account type affects investment risk and the amount of that risk borne by the government via taxes. Tabor says that the government bears some of the investment risk in a tax-exempt account. Sills says the government bears some of the risk in a tax-deferred account before withdrawals occur. With respect to those assertions:

A) both Tabor and Sills are correct.

B) Tabor is correct and Sills is incorrect.

C) both Tabor and Sills are incorrect. ✓

Explanation

If the investment is held in a tax-exempt account, then the investor bears all the investment risk because there is no tax paid to the government for investment gains, nor tax loss benefits provided for investment losses. That is also correct for tax-deferred accounts before withdrawals occur because, even though the government taxes the future accumulation, the variability of returns during the deferral period is not reduced by taxes levied at the time of withdrawal. During the deferral period, there are no taxes levied by the government, so the government does not bear any of the risk before the withdrawals occur.

(Module 5.1, LOS 5.b)

Question #6 of 18

Question ID: 1587537

The rebalancing ranges for a taxable portfolio, compared to a tax-exempt portfolio with similar risk characteristics, are typically:

A) narrower, because it takes smaller asset class movements to shift the risk profile of a taxable portfolio. ✗

B) narrower, because after-tax volatility is lower than pretax volatility. ✗

C) wider, because after-tax volatility is lower than pretax volatility. ✓

Explanation

A taxable portfolio must pay taxes when the portfolio is profitable (or receive a tax loss carryforward when the portfolio has a loss). The effects of reduced returns in profitable periods and reduced losses in loss periods demonstrates that after-tax volatility is lower than pretax volatility.

For taxable portfolios, lower after-tax volatility means that larger asset class movements are required to affect the risk profile of a portfolio, so the rebalancing ranges can be wider.

(Module 5.1, LOS 5.b)

Question #7 of 18

Question ID: 1587543

Peter and Cornelia Hargrave, both age 60, have a $2.5 million investment portfolio. The Hargraves want to retire in five years, and as a result, they use a strategic asset allocation of 55% equities, 40% bonds, and 5% cash. The portfolio is expected to provide income for 20 years after retirement, and it is invested among two equity portfolio managers and a fixed income manager. Last week, the Hargraves learned that one of the equity portfolio managers has taken on extremely risky derivatives positions generating substantial losses that wiped out $500,000 of the $2.5 million portfolio. Given the change in portfolio value, the Hargraves should reduce the exposure under the strategic asset allocation to which asset class?

A) Bonds. ✗

B) Cash. ✗

C) Equities. ✓

Explanation

Given the significant reduction in the portfolio value (20%) and only a modest amount of time before retirement, the Hargraves should reduce exposure (in terms of the percentage of the *remaining* portfolio assets) to riskier assets, such as equities, and increase exposure to less risky assets, such as fixed income. The fixed-income assets should be duration matched to meet the time to retirement, income needs at retirement, and life expectancy.

(Module 5.2, LOS 5.c)

Question #8 of 18

Question ID: 1587535

Julia Adamson is 77 years old, with a life expectancy of nine years. Adamson has told her portfolio manager that her portfolio needs to fund three buckets that each allow her to maintain her current standard of living: (1) from now until age 80, (2) age 80 to 90, and (3) age 90 to 100. Which of the three buckets should allow for the highest allocation to risky investments?

A) Bucket 3 (age 90 to 100). ✓

B) Bucket 1 (until age 80). ✗

C) Bucket 2 (age 80 to 90). ✗

Explanation

Given Adamson's life expectancy of nine years, Buckets 1 and 2 have the greatest priority. Investments allocated to those two buckets should be more conservative, with a greater emphasis on liquidity. Bucket 3 is at least four years outside of her life expectancy range; therefore, it can be invested in higher risk investments to ensure asset growth for the longer term, should she exceed her nine-year life expectancy.

(Module 5.1, LOS 5.a)

Question #9 of 18

Question ID: 1587533

A pension fund is searching for an optimal portfolio by comparing the present value of contributions and contribution risk as measured by a 95% confidence interval that contributions will not exceed a given threshold. Which of the following portfolios would be the *most efficient* under these criteria?

A) PV contributions: $69 million; contribution risk: $390 million.

B) PV contributions: $60 million; contribution risk: $385 million.

C) PV contributions: $65 million; contribution risk: $392 million.

Explanation

The best choice will minimize contribution risk, which is the chance of having to increase contributions, and will minimize the expected present value of future contributions.

(Module 5.1, LOS 5.a)

Question #10 of 18

Question ID: 1587544

The strategic asset allocation, current allocation, and upper and lower policy limits for a retirement portfolio are shown in the following table.

Asset Class	Current Weight	SAA	Upper Limit	Lower Limit
Fixed income	35%	40%	45%	35%
Domestic equity	60%	50%	60%	40%
Real estate	5%	10%	15%	5%

The portfolio manager takes a discretionary approach to tactical asset allocation. She follows various indicators that she believes are useful in evaluating specific asset classes. Her current observations and interpretations are as follows:

- Equity brokers have reported that margin borrowing levels increased dramatically over the last two months and now sit at their highest levels for 15 months—a bearish sign.
- Real estate indices have started to rise, suggesting the market may be "heating up"—a bullish sign.
- Indications from the Federal Reserve are that interest rates will rise several times in the next 12 months—a clearly obvious sign.

Which of the following tactical allocation shifts is the manager *most likely* to make?

A) Increase the weighting to real estate.

B) Increase the weighting to domestic equity.

C) Decrease the weighting to fixed income.

Explanation

She is bullish on RE, and RE is at the bottom of her range; she should increase the allocation, if possible. Admittedly, RE is a less liquid asset. She is bearish on equity and would not want to increase the weight. Increasing interest rates is bearish for bonds, but she is already at the bottom of the range. Avoid convoluted arguments that to change one weight, she must do something else as well. That is not a direct answer to the question, was not asked, and we'd be guessing what that other action is. She may already have some cash assets.

(Module 5.2, LOS 5.d)

Question #11 of 18

Question ID: 1587547

The investment committee of a large-cap fund consists of 12 members. During committee meetings, one of the senior members is frequently the first one to speak and speaks with strong confidence, which tends to bias the other committee members' views. Which behavioral bias is *most likely* to occur within the committee?

A) Recency bias. ✗

B) Framing bias. ✓

C) Illusion of control bias. ✗

Explanation

Framing bias occurs when decision-making is affected by the way information is presented. For example, in committee settings, a member who typically speaks first and with authority can bias the views of others.

The illusion of control is a bias in which individuals overestimate their ability to control events. Recency (or representative) bias causes investors to give more importance to recent events or data, rather than older events or data. None of those biases is explicitly demonstrated within the fact pattern.

(Module 5.3, LOS 5.e)

Question #12 of 18

Question ID: 1587539

If an investment is held in an account that is taxed annually, the government bears:

A) none of the investment risk. ✗

B) some of the investment risk. ✓

C) all of the investment risk. ✗

Explanation

If the investment returns are taxed solely as income at the tax rate t and the pretax standard deviation of returns is S, then the investor's after-tax risk is $S \times (1 - t)$, and the government bears a portion of the risk.

(Module 5.1, LOS 5.b)

Question #13 of 18

Question ID: 1551744

On a graph where the risk is on the horizontal axis and the returns are on the vertical axis, the existence of taxes on investment returns would probably shift the mean-variance optimization portfolio:

A) down and to the right. ✗

B) down only, and there would not be a shift left or right. ✗

C) down and to the left. ✓

Explanation

Taxes lower returns, but they also shift some of the investment risk to the government.

(Module 5.1, LOS 5.b)

Question #14 of 18

Question ID: 1587546

Goals-based investing is most directly useful to counter:

A) the illusion of control. ✗

B) loss aversion. ✓

C) representativeness. ✗

Explanation

The answer is loss aversion based specifically on what the reading says. You have taken at least two CFA exams and know such picky questions are unusual. Exhibit good judgment, give your best guess, and move on.

(Module 5.3, LOS 5.e)

Question #15 of 18

Question ID: 1587545

Kevin Collison has several different places with assets in which he manages as separate accounts: his checking account, which he uses for short-term cash and emergency needs, his 401(k) account for retirement, and his children's college fund. In his 401(k), he owns a small-cap stock in a company that makes catheters to be used in experimental cancer treatments, in which the catheter is used to deliver cancer-killing drugs too hard to reach tumors in the body. The stock has recently taken a downturn in price due to the FDA not approving their most recent catheter. As a result of the downturn, Collison purchases more of the stock in hopes of recouping his losses at some future time. Collison's management of his portfolio is indicative of:

A) mental accounting and loss aversion.

B) money illusion and fear of regret.

C) asset allocation and pyramiding.

Explanation

Collison is exhibiting mental accounting by having separate accounts for his assets, where he manages each account separately for a specific purpose and does not manage his assets as a complete portfolio for asset allocation purposes. He is also exhibiting loss aversion by not accepting the loss on the stock, which is also leading to risk-seeking behavior by buying more of the losing stock.

(Module 5.3, LOS 5.e)

Question #16 of 18

Question ID: 1606709

Dan Vustings, 51, is a senior account executive at an advertising agency in San Francisco. He currently has sufficient capital in his two investment portfolios to retire in 14 years. His capital is evenly split between a taxable portfolio, which he uses to fund current consumption goals and a tax-deferred retirement account. Vustings received advice from an investment manager last year who suggested the following asset allocation:

- High-yield bonds: 15%
- High-dividend-yield equities: 50%
- High growth equities: 35%

Vustings has the same allocation in both the taxable and the retirement account. Which of the following allocations would *most likely* improve the tax efficiency of the portfolios, ignoring rebalancing or withdrawal penalties?

A) Increase the allocation to high-yield bonds in the taxable account, and increase the allocation to high growth equities in the retirement account.

B) Increase the allocation to high-yield bonds in the retirement account, and increase the allocation to high growth equity investments in the taxable account. ✓

C) Increase the allocation to high-dividend yield equities in the taxable account, and increase the allocation to high growth equities in the retirement account. ✗

Explanation

Assets subject to the highest rates of tax should be first allocated to tax-advantaged accounts. Interest on high-yield bonds is typically taxed at a higher rate than dividend income—which, in turn, is taxed at a higher rate than capital gains. Vustings should, therefore, allocate high-yield bonds first to the retirement account, and leave the high growth equity in the taxable account.

(Module 5.1, LOS 5.b)

Question #17 of 18

Question ID: 1587538

Ruby Pascal is an institutional money manager managing a $200 million portfolio. The cost basis of each of five selected assets from the portfolio is above market value, and Pascal decides to sell those assets. The *least likely* reason for the sale is:

A) to realize capital gains to generate profits for the portfolio. ✓

B) to realize capital losses against current or future realized capital gains. ✗

C) to engage in tax loss harvesting. ✗

Explanation

Before a sale, when the cost basis of an asset *exceeds* its market value, the asset has an unrealized *loss*. The primary reason for selling assets with unrealized losses is to engage in tax loss harvesting. Tax loss harvesting involves realizing capital losses against current or future realized capital gains to reduce the portfolio's overall tax liability.

(Module 5.1, LOS 5.b)

Question #18 of 18

Question ID: 1551748

In assessing the effectiveness of the risk-adjusted tactical asset allocation (TAA) portfolio against the strategic asset allocation (SAA) portfolio, an investment manager should compare the risk-adjusted TAA portfolio's risk and return with the:

A) risk and return of the portfolios along the SAA portfolio’s efficient frontier. ✓

B) Sharpe ratio of the SAA portfolio.

C) information ratio of the SAA portfolio.

Explanation

The risk-adjusted TAA portfolio's effectiveness is best assessed by plotting its realized risk and return against the risk and return of the portfolios along the SAA portfolio's efficient portfolio. Even if the TAA portfolio had better Sharpe and information ratios than the SAA portfolio, it could still be less optimal than other portfolios along the SAA's efficient frontier.

(Module 5.2, LOS 5.d)